HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed bitwise accurate **C-Model** simulations for next-generation GPU architectures, translating design specifications into functional C++ code.
- Engineered **C++ test suites** to perform functional verification and validation against C-Models and RTL simulations for complex graphics pipelines.
- Executed **conformance tests**, stress tests, and random tests on GPU and Compute systems, identifying and resolving stability issues.
- Contributed to GPU architecture design by developing algorithms for functional blocks and verifying their correctness in collaboration with architects and designers.
- Wrote **Python scripts** to automate data extraction and analysis from simulation logs, improving the efficiency of test result processing by approximately 30%.
- Participated in **Agile development methodology** ceremonies, including sprint planning and daily stand-ups, to ensure timely delivery of modeling features.
- Managed **Linux-based development environments**, including shell scripting for build automation and log analysis to support debugging efforts.

SKILLS

GPU Architecture – Computer Architecture, Graphics Pipelines, GPU Modelling, C-Model, RTL Simulation, Conformance Tests, Stress Tests, Compute Systems

Programming Languages – C++, C, Python, Perl

Development Tools – Git, Jenkins, Jira, Confluence

Methodologies – Agile, Scrum

Cloud & Containerization – AWS, Docker

Databases & Scripting – SQL, Shell Scripting

PROJECTS

GPU Simulation & Verification Framework

- Developed a **C++ simulation framework** for a new GPU architecture, enabling bitwise accurate modelling of core functional blocks.
- Implemented a suite of **regression tests** in C++ to validate the functional correctness of the C-Model against architectural specifications, achieving 95% test coverage.
- Integrated **Python scripting** for automated test execution and result aggregation, reducing manual effort in the verification cycle by an estimated 40%.

Graphics Pipeline Performance Analysis Tool

- Created a **Python-based analysis tool** to parse and visualize performance metrics from graphics rendering simulations.
- Designed and implemented algorithms to identify **performance bottlenecks** in the compute pipeline, providing actionable insights for optimization.
- Utilized **SQL** to store and query historical performance data, enabling trend analysis over multiple simulation runs.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)